Lesson Plan · Monday, November 16

Algebra I · Mr. Pandey · Module 3, Topic 1, Lesson 3: Modeling Linear Inequalities (Day 3 of 3) · essentials only (✪)

Objectives (students will…)

- Solve inequalities with variables on both sides.
- Solve inequalities that need the distributive property.
- Graph each solution set on a number line.

TEKS

- **A.5B** solve linear inequalities, including distributive property and variables on both sides
- Process: A.1F

Key terms

- (review) linear inequality · solve an inequality

Materials

- Slides: m3t1-lesson3-modeling-linear-inequalities (.html/.pptx) · publisher slides with answers: BLMathematics_AlgI_M3_T1_L3_LessonSlides_ENG_with_answers.pptx
- Bell work: BELL WORK/T1-Bell-Work-Mon-11-16.pdf + key (online: m3t1-lesson3-bellwork-day3)
- Exit ticket: EXIT TICKET/T1-Exit-Ticket-Mon-11-16.pdf + key (written like the End of Topic 1 test)
- Video: videos/M3T1-Lesson3-Linear-Inequalities.mp4 · IEP and 504 versions of all of these: IEP/, Section 504/, and the -IEP / -504 PDFs
- Student Edition pp. 380–382 (Activity 3.4, Talk the Talk)
- Topic 1 Self-Reflection SE pp. 387–388 (for homework)
- Whiteboards

Bell work (on the board)

- What is the solution set of $-4x > 12$? D. $x < -3$
- What is the solution set of $2x + 3 \leq x - 1$? F. $x \leq -4$
- Which inequality means the same as $-6 \leq x$? B. $x \geq -6$
- What value of x makes $3(x - 2) = 9$ true? H. 5

Agenda (60 minutes)

Time	Activity	What to do / say
0–5	Bell work	Q3 sets up Kai's answer in 3.4.
5–30	✪ Activity 3.4: Solving Other Linear Inequalities · SE pp. 380–381	$5x + 2 \geq 3x - 10$: Victoria moves x left → $x \geq -6$; Kai moves x right, divides by −2 → $-6 \leq x$, same answer. Q2: same as equations, but watch the sign. $-4(x - 6) < 22$: Aaliyah distributes first, Ava divides by −4 first → both $x > ½$. Q4: distributing works the same way.
30–50	✪ Talk the Talk: It's About Solutions, More or Less · SE p. 382	Solve and graph: a $-⅔x \geq 7$ $x \leq -21/2$ b $32 > 23 - x$ $x > -9$ c $15 - 4x \leq 3x - 6$ $x \geq 3$ d $2(x + 6) < 14 + 4x$ $x > -1$ Q2 if A < B: A + C < B + C always; AC > BC only when C < 0; −A < −B never.
50–58	Exit ticket	1. What is the solution set for $3(4 - x) > -2x + 10$? $x < 2$ 2. Solve and graph: $1.5m + 4 \geq -2(m - 9)$ $m \geq 4$, closed circle at 4, shaded right 3. Solve: $-2(x + 1) \geq 8$ $x \leq -5$ IEP/504: read aloud on request, properties/notes card, extra time.
58–60	Wrap up	Essential Question (SE p. 371). Homework: Lesson 3 Assignment Practice 3a–d (SE p. 385) + Topic 1 Self-Reflection. Review tomorrow, test Wed 11/18.

Check for understanding

- Whiteboards: "Which side do you put the x on to avoid flipping?" the side with the bigger coefficient
- Thumbs: "$2(x + 6) < 14 + 4x$: the sign flips." Down if you keep x on the right
- Activity 3.4 Q1–4 are the TE's check points.

After class

- Use exit tickets + Self-Reflection to group students for Tuesday's review.
- Practice 3 answers: $x > -1$, $x < 7$, $d \leq 11$, $k \leq -24$.
- Missed both exit items → Skills Practice Set D.

3.4 and Talk the Talk are starred. That finishes Lesson 3.